

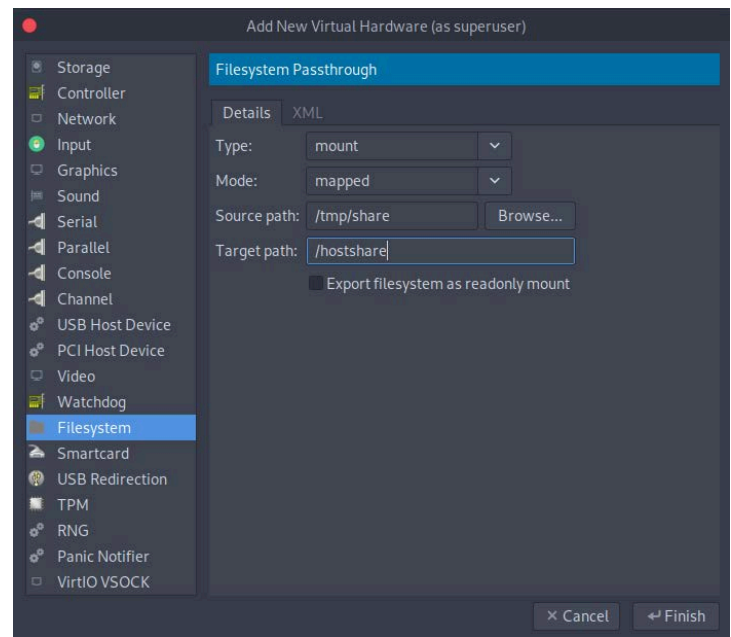
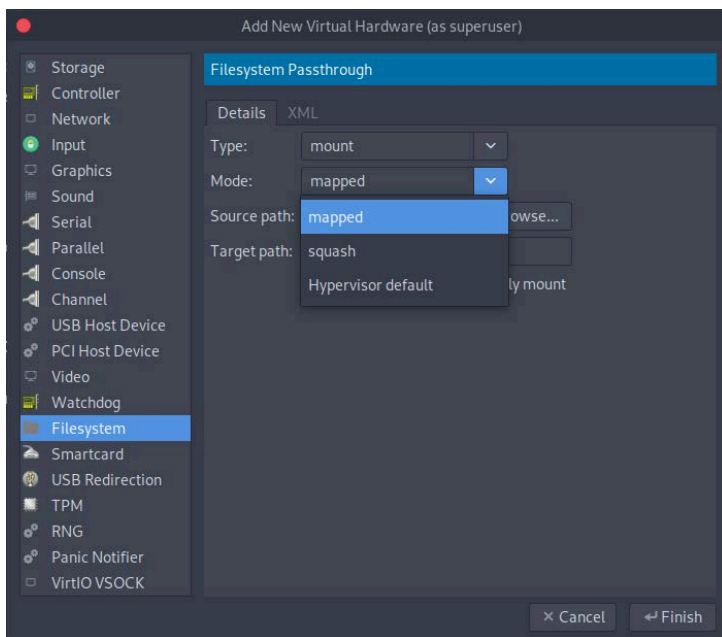
#1 Share Host Directories/Files with the Guest (Linux) filesystem passthrough

Filesystem Passthrough, Managed (Graphically), using virt-manager, for linux Guests

This assumes you've already installed a guest with virt-manager or virt-install and it's shut off. Start virt-manager, and open your VM by double clicking on it. Click the virtual hardware details.

Click add Hardware, and choose Filesystem.

Chose the mode, we chose mapped (see QEMU wiki for details on the security models), the source (we are sharing /tmp/share), and the target (we called this /hostshare).



Now, start the VM. **In the guest** we need to mount the 9p filesystem from the host using the virtio transport. The mount_tag is used to identify the host's share.

mkdir /tmp/host_files (create-directory-of-mount-point-for-shared-folder-on-guest) if not already exists

mount -t 9p -o trans=virtio (target path; shared-external-folder ex: /hostshare) (mount point-on-guest ex: /tmp/host_files)

That's it...now we can read/write files in that directory from either the host or the guest. (Note: likely to hit some issues w/ privileges since libvirt runs guests unprivileged. so execute as sudo.

With QEMU's 9pfs you can create virtual filesystem devices (virtio-9p-device) and expose them to guests, which essentially means that a certain directory on host machine is made directly accessible by a guest OS as a pass-through file system by using the 9P network protocol for communication between host and guest, if desired even accessible, shared by several guests simultaneously.

You can mount the shared folder using: (Within the Guest)

```
mount -t 9p -o trans=virtio [mount tag] [mount point]
```

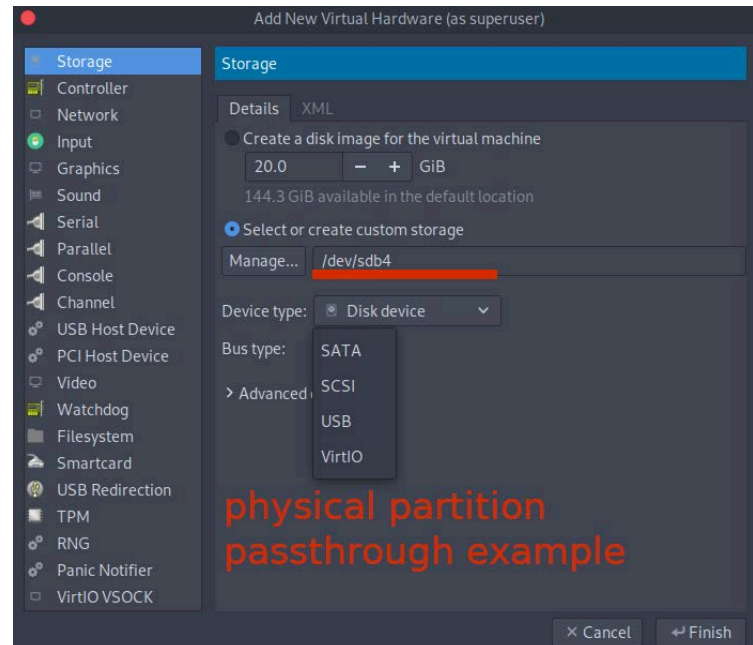
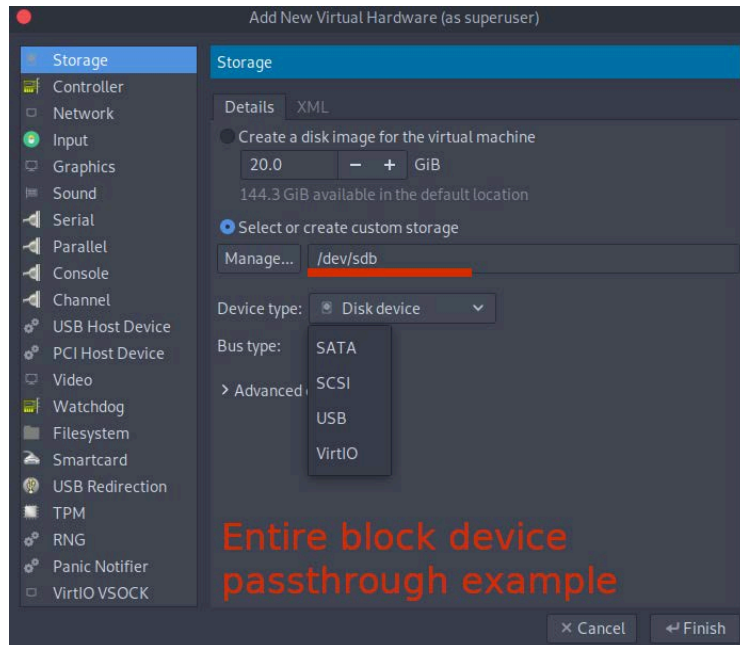
```
$ sudo mount -t 9p -o trans=virtio (target-path-on-GUEST-ex: /mount-test) (desired-mount-point-on-GUEST-ex: /mount-external)
$ sudo mount -t 9p -o trans=virtio /source-on-guest /target-on-guest
```

***** target path must be in the root / of the guest os**

#2 ADDING HARD DRIVES AND OTHER BLOCK DEVICES TO A GUEST (Passthrough Entire Physical-Partitions & Physical-Drives)

Adding Physical Block Devices to Linux Guests:

you can passthrough physical partitions or entire physical drives to a linux guest .



The devices or partitions can be assigned in different ways:

- `/dev/disk/by-uuid/<uuid of device>`
- `/dev/disk/by-partuuid/<uuid of partition>`
- `/dev/disk/by-label/<label of device>`
- `/dev/disk/by-id/<id of partition>`
- `/dev/disk/by-path/<path of partition>`

* you can also assign the devices/partitions parameters manually by editing the xml file

Passthrough Entire block devices to a Windows Guest

- You must **install virtio drivers for windows!** :Highly recommended to download virtio-win iso from fedora
- you can only passthrough entire block devices to windows guests (no partitions, only entire devices)

*****note: it is perfectly fine to passthrough an entire device to a guest that is running on that same block device